

Predictive Simulation Framework Based Muscle Effort Replication During Walking on Different Terrains with Ankle Exoskeleton

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Abstract—Existing exoskeleton-related predictive simulation studies primarily focused on evaluating the consistency between experimental and simulation results. Most of these studies examined kinematic adaptations and metabolic energy expenditure changes under exoskeleton assistance during level-ground walking. Research examining muscle activation and muscle force changes with experimental findings, as well as those considering terrain factors, remains relatively limited. This study presents a predictive simulation framework to evaluate the biomechanical effects of exoskeleton assistance, with particular emphasis on muscle effort analysis (muscle activation and muscle force) in different terrains. Experiments were conducted under conditions corresponding to the simulations, and the consistency between experimental and simulation results was comprehensively evaluated to validate the efficacy of the proposed framework. Through qualitative and quantitative analyses, this study demonstrates that the proposed predictive simulation framework can reasonably predict the trends of changes in kinetics, plantar flexor muscle activation, and plantar flexor muscle force under exoskeleton assistance across different terrains.

Index Terms—Predictive simulation, exoskeleton assistance, muscle effort, multi-terrain walking.

I. INTRODUCTION

EXOSKELETON demonstrate significant potential in enhancing human motor performance and reducing musculoskeletal burden, with applications spanning rehabilitation, industrial assistance, and military enhancement [1–5]. As these technologies advance, optimizing exoskeleton design and

control strategies has become a crucial focus in biomechanics and rehabilitation engineering research [6–9]. Innovations in human-robot interaction optimization and temporal assistance modulation, particularly through human-in-the-loop optimization methods, have substantially reduced walking metabolic costs [10]. However, these approaches still require time-intensive testing protocols, often demanding several hours of walking time that may need to be distributed across multiple sessions [11]. Predictive simulation avoids experimental gait data requirements by solving optimal control problems, highlighting its importance in computational evaluation of exoskeleton controller design, parameter settings, and individual biomechanical variations [12–14]. This approach promises to significantly shorten design cycles and reduce the burden of human testing.

In recent years, predictive simulation has emerged as a growing research focus in assisting exoskeleton design and control [15]. By simulating human walking behavior under exoskeleton assistance, predictive simulation enables observation of human adaptation and supports optimization of exoskeleton control parameters [12, 14, 16–18]. However, existing exoskeleton-related predictive simulation studies primarily focused on validating the consistency between experimental and simulation results regarding metabolic energy expenditure and kinematic adaptations induced by exoskeleton assistance during level-ground walking [14, 17, 19, 20]. Research examining muscle activation and muscle force changes with experimental findings, as well as those considering terrain factors, remains relatively limited.

Some exoskeleton designs focus on reducing metabolic energy expenditure, while others are dedicated to alleviating muscle load and associated injury risks. Ankle exoskeletons can reduce the activation levels of postural control muscle groups such as the soleus muscle by providing plantarflexion torque assistance, while simultaneously reducing muscle forces in primary propulsive muscles such as the gastrocnemius, thereby effectively relieving muscle fatigue and reducing exercise-related injury risks [21]. Predictive simulation studies of muscle activation and muscle force under exoskeleton assistance may contribute to evaluating the effectiveness and optimizing the performance of exoskeleton systems. Movement in natural environments involves diverse terrain, including level ground, uphill, and downhill conditions, which produce different effects on lower limb biomechanical characteristics. In contrast to level walking, ankle plantar flexor muscle activation increases significantly during uphill walking, whereas during downhill walking, ankle muscle demands decrease while mechanical loading at the

This work was supported by the National Natural Science Foundation of China (No. 52375001, No. 52475001, and No. 524B2044) and the Beijing Natural Science Foundation (No. L254009). (Xiaolin Dai and Ming Xu contributed equally to this work.) (Corresponding author: Zhihao Zhou)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Local Ethics Committee of Peking University First Hospital, China (under Application No. #2021-scientific research-428).

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knee and hip joints intensifies [22]. These terrain differences result in drastically different force distributions across body joints, requiring humans to make corresponding biomechanical adjustments to maintain balance and efficiency when walking on different slopes. These biomechanical changes induced by terrain directly affect exoskeleton assistance outcomes [23]. Predictive simulation research on multi-terrain walking with exoskeleton assistance may also facilitate the assessment and enhancement of exoskeleton-assisted locomotion.

Predictive simulation studies of exoskeletons typically employ qualitative analysis of experimental and simulation results to validate the authenticity of simulated movements [15]. Some studies have conducted quantitative analysis of kinematic adaptations and metabolic energy expenditure changes under exoskeleton assistance [17, 19]. Conducting quantitative analysis by comparing experimental results with simulation results, including kinematics, kinetics, and muscle effort (muscle activation and muscle force), could facilitate more comprehensive validation of the consistency between simulated exoskeleton-assisted movements and actual movements.

This study proposes a predictive simulation framework for evaluating the biomechanical effects of exoskeleton assistance, with special focus on muscle effort (muscle activation and muscle force) analysis and extension to multi-terrain walking scenarios. The framework employs a reflex-based control policy optimization method to investigate the effects of unilateral ankle exoskeleton assistance on ankle joint biomechanical characteristics during walking across different terrains, including kinematics, kinetics, and muscle effort. The accuracy of this predictive simulation framework is validated through both qualitative and quantitative analyses of simulated results compared with their paired experimental results. This study might provide a theoretical basis for more precise exoskeleton design and control optimization in the future.

II. PREDICTIVE SIMULATION FRAMEWORK

A. Predictive Simulation Platform

Predictive simulations can be generally categorized into two main approaches: trajectory optimization and control policy optimization [15]. Trajectory optimization methods are computationally efficient and readily implementable, but accurately reproducing human gait typically requires iterative parameter tuning or computationally intensive bilevel optimization [24]. In contrast, reflex-based neuromuscular control policy optimization approaches have received limited attention, especially in the context of human-exoskeleton interaction. By capturing the inherent robustness of the neuromusculoskeletal system to external assistance and disturbances [25, 26], these methods provide a promising avenue for investigating neuromuscular adaptation to exoskeleton assistance across diverse terrains, motivating their adoption in the present study.

This study employs neuromuscular skeletal modeling for forward dynamic simulation to investigate human gait characteristics with and without exoskeletal assistance. As shown in Fig. 1, the simulation framework, based on SCONE (v2.4.0) and Hyfydy platforms, which implements the aforementioned reflex-based neuromuscular control policy optimization, enables the prediction of kinematics and kinetics during a

20-second continuous gait cycle [27]. The musculoskeletal model employed in this study is the H0918 two-dimensional musculoskeletal model (height: 1.68 m, body mass: 74.53 kg), consisting of eight bodies: head-and-trunk (HAT), pelvis, and three segments for each leg (thigh, shank, and foot). This configuration incorporates nine degrees of freedom, all constrained to the sagittal plane: pelvis-ground vertical and horizontal displacements, pelvic tilt, and three degrees of freedom for hip, knee, and ankle joints in each leg. Actuation is provided by nine Hill-type muscles per leg, encompassing the iliopsoas, gluteals, rectus femoris, hamstrings, biceps femoris short head, vastus, gastrocnemius, soleus, and tibialis anterior. Foot-ground contact is simulated using two visco-elastic Hunt-Crossley contact spheres at the heel and toe of each foot.

The predictive simulation framework integrates four interacting modules: the neuromuscular controller, the exoskeleton controller, the musculoskeletal system, and the CMA-ES optimizer. During each forward dynamic step, the musculoskeletal system outputs the current sensory states, including muscle fiber length, contraction velocity, muscle force, joint angles, and GRF. The GRF is used by both controllers to estimate the current gait phase through a step detection mechanism, ensuring that the neuromuscular controller and the exoskeleton controller operate on a synchronized gait timeline. Based on this shared phase information, the neuromuscular controller generates muscle excitations through reflex laws, while the exoskeleton controller independently produces a phase-dependent assistive plantarflexion torque profile. The muscle excitations drive the Hill-type muscle actuators, and the assistive torque is applied at the unilateral ankle joint as an external input to the musculoskeletal system. The combined effect updates the musculoskeletal states at the next time step, and the updated states are fed back to the neuromuscular controller, thereby closing the sensorimotor loop.

A reflex-based neuromuscular controller with parameterization is used to generate gait patterns, following the approach of Geyer and Herr [28]. Within this control architecture, the sensory feedback signals, muscle fiber length, contraction velocity, and muscle force, are continuously obtained from the musculoskeletal model during forward dynamic simulation and fed back to the neuromuscular controller (denoted as the "Sensory Feedback loop in Fig. 1). These state-dependent signals serve as inputs to the reflex-based control laws, where muscle excitations are computed as weighted combinations of length-, velocity-, and force-related feedback terms, modulated by gait-phase-dependent reflex gains. The resulting muscle excitations drive the Hill-type muscle actuators to generate joint torques and GRF, thereby updating the musculoskeletal states at the subsequent time step. Through this iterative process, a closed-loop neuromuscular control system is established, enabling the model to autonomously generate stable gait patterns and to adapt to variations in terrain conditions and exoskeleton assistance without relying on prescribed experimental kinematics.

It should be emphasized that the neuromuscular controller and the exoskeleton controller are not hierarchically nested but operate in parallel. The exoskeleton controller is dedicated to producing a stable and consistent assistive torque profile that is delivered to the ankle joint of the musculoskeletal

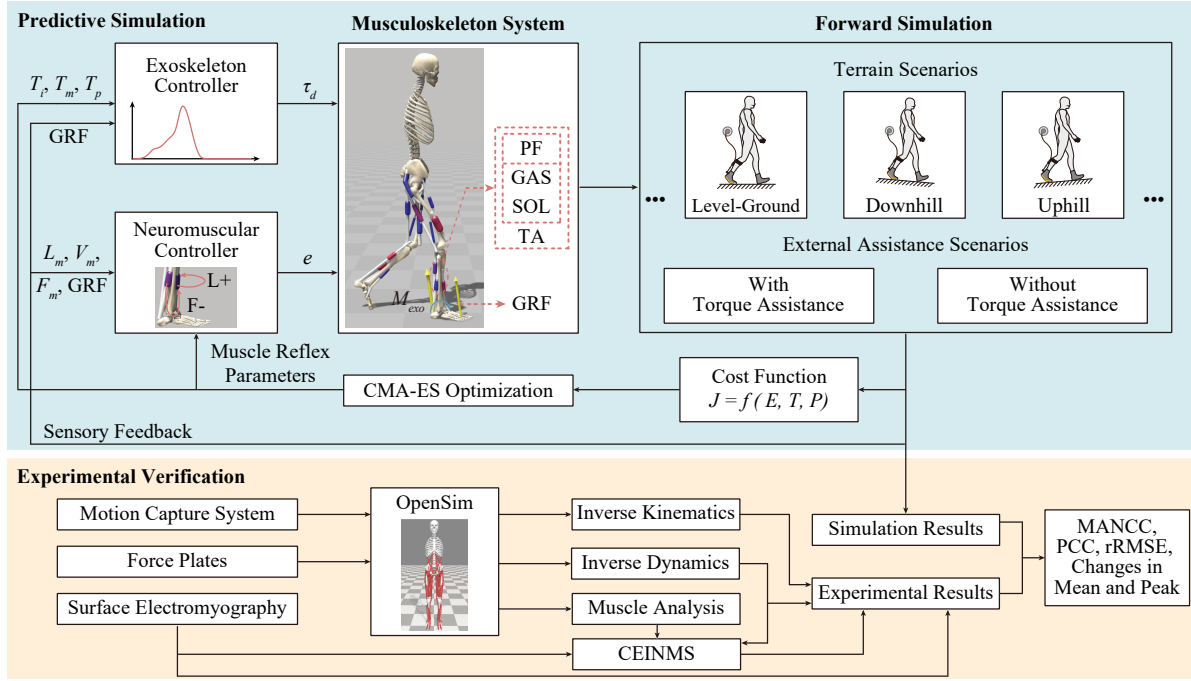


Fig. 1. Predictive simulation framework and experimental verification. The framework consists of two main parts: predictive simulation and experimental verification, aimed at evaluating changes in kinematics, kinetics, and muscle effort (muscle activation and muscle force) during multi-terrain walking with exoskeleton assistance. The predictive simulation simulates human movement by setting terrain parameters, walking speed, and exoskeleton control parameters. The sensory feedback loop continuously transmits muscle fiber length (L_m), contraction velocity (V_m), muscle force (F_m), and ground reaction force (GRF) from the musculoskeletal system to the neuromuscular controller, which generates muscle excitations (e) through reflex laws to drive forward dynamic simulation, forming a closed-loop reflex control scheme. The consistency between experimental and simulation results is evaluated using several analytical methods, including the maximum normalized cross-correlation coefficient (MANCC), Pearson correlation coefficient (PCC), relative root mean square error (rRMSE), and percentage changes in mean and peak. The neuromuscular controller and the exoskeleton controller operate in parallel and share the gait phase derived from GRF. The exoskeleton controller delivers the prescribed assistive torque (τ_d) to the ankle joint of the musculoskeletal model, while the neuromuscular controller adapts its muscle excitations passively through the sensory feedback loop in response to the resulting changes in musculoskeletal states.

model. The neuromuscular controller does not receive the assistive torque command directly; instead, it perceives the mechanical consequence of the assistance only through the altered sensory feedback (e.g., reduced plantar flexor muscle force and modified ankle kinematics) and adapts its excitations accordingly through reflex laws. Through this mechanism, the two controllers reach a passive adaptive balance during forward dynamic simulation: the exoskeleton consistently delivers the prescribed assistance, while the neuromuscular system spontaneously reorganizes its activation pattern in response to the resulting changes in muscle and joint states. This decoupled architecture preserves the biological plausibility of the reflex-based control and allows neuromuscular adaptation to emerge naturally from state-dependent feedback rather than from prescribed coordination with the exoskeleton.

The reflex-based controller mathematically models muscle excitations during each gait phase to facilitate model walking. This control scheme has been extensively validated across multiple studies, including applications in exoskeletons, prosthetics, and gait disorder analysis. [12, 25, 29–31]. Controller parameter optimization employs the covariance matrix adaptation evolution strategy (CMA-ES) to minimize a cost function [32]. The optimized parameters comprise neural controller reflex parameters (Table I of [28]), the GRF threshold for step detection, and initial postural conditions including pelvic tilt

and bilateral hip, knee, and ankle angles [33]. The population size used in the CMA-ES optimization is set to 16 [34]. The cost function is

$$J = 0.1E_{\text{CoT}} + 2000E_{\text{muscle}} + 0.02|T_{\text{lim, knee}}| + 0.1P_{\text{RoM, ank}}^2 + 10P_{\text{GRF peaks}} + 100(P_{\text{vel}} + P_{\text{tend}} + P_{y_{\text{CoM}}}), \quad (1)$$

where E_{CoT} indicates metabolic energy [35], E_{muscle} indicates muscle activation [27], knee joint stop torques $T_{\text{lim, knee}}$, and penalties P for undesired behaviors: P_{vel} (walking speeds < 1.0 m/s), P_{tend} (simulation termination < 10 s), $P_{y_{\text{CoM}}}$ (center of mass position < 0.85 m), $P_{\text{RoM, ank}}$ (ankle angles $> 60^\circ$ dorsiflexion/plantarflexion), and $P_{\text{GRF peaks}}$ (GRF > 1.4 times body weight). See SCONE documentation for details [36].

Each weight coefficient in the cost function reflects both the numerical scale of the associated term and its relative priority during optimization. The metabolic energy term E_{CoT} and the muscle activation term E_{muscle} serve as the primary performance-oriented objectives, encouraging the optimizer to converge toward an energy-efficient gait pattern with physiologically reasonable muscle effort. The relatively small coefficient (0.1) assigned to E_{CoT} reflects the fact that the cost of transport typically takes values on the order of several J/(kg·m), whereas the coefficient (2000) assigned to E_{muscle}

compensates for the small numerical magnitude of squared muscle activations (bounded between 0 and 1) and ensures that muscle effort minimization contributes meaningfully to the overall objective. The knee stop torque term $T_{\text{lim,knee}}$, weighted by 0.02, suppresses excessive knee hyperextension impacts that may otherwise emerge from unconstrained optimization, while its small weight prevents this term from dominating the search direction. The ankle range-of-motion penalty $P_{\text{RoM,ank}}^2$, weighted by 0.1 and squared to provide smooth gradient information, discourages physiologically unrealistic ankle excursions. The ground reaction force peak penalty $P_{\text{GRF peaks}}$, weighted by 10, prevents abnormally large impact peaks that would compromise the biomechanical plausibility of the simulated gait. The remaining stability-related penalties, P_{vel} , P_{tend} , and P_{yCoM} , are jointly weighted by 100 and serve as feasibility constraints that enforce the prescribed walking speed, sufficient simulation duration, and an upright center-of-mass height, respectively. Their relatively large weight ensures that feasibility constraints are satisfied with high priority before the optimizer further refines the gait pattern in terms of energetic and muscular efficiency. This hierarchical weighting strategy balances task feasibility, biomechanical plausibility, and energetic efficiency within a single scalar objective.

The cost function J governs the optimization of the neuromuscular controller parameters by the CMA-ES algorithm, including the reflex gains and offsets associated with length-, velocity-, and force-related feedback for each muscle, the GRF threshold for step detection, and the initial postural conditions (pelvic tilt and bilateral hip, knee, and ankle angles). These parameters jointly shape the metabolic energy term E_{CoT} , the muscle activation term E_{muscle} , the knee stop torque term $T_{\text{lim,knee}}$, and the stability-related penalties (P_{vel} , P_{tend} , P_{yCoM} , $P_{\text{RoM,ank}}$, $P_{\text{GRF peaks}}$). Together, these terms drive the optimization toward a stable and energy-efficient gait pattern.

The cost function weights adopted in this study follow the established settings of the SCONE framework and have been validated in prior predictive simulation studies that employed the same reflex-based neuromuscular controller and CMA-ES optimizer [27, 36–38]. In particular, a dedicated numerical sensitivity analysis on this set of weights was reported in a recent study [38], in which the largest weight was halved and the two smallest weights were doubled in order to substantially reshape the objective landscape. Under this perturbation, the predicted kinematic and kinetic waveforms, as well as the rankings of the key biomechanical metrics, remained preserved across all walking conditions. This stability further supports the rationale that the chosen weights are determined primarily by the numerical scale and physical role of each term, rather than by fine-grained empirical tuning specific to a particular walking task or terrain. Accordingly, the same weight selection is retained in the current study.

B. Simulations for Different Scenarios

The generic musculoskeletal model (H0918 two-dimensional musculoskeletal model) was scaled based on each subjects body weight and marker calibration poses

to create personalized models. The simulation scenarios encompass three terrain conditions: level-ground, 5-degree uphill, and 5-degree downhill, with a minimum walking speed threshold set at 1.0 m/s. A plantarflexion moment profile resembling the positive biological ankle joint moment is generated beginning in early stance. This assistive technique, termed ankle moment inspired technique (AMIT), is designed to reduce biological joint moment and power throughout the stance phase [39, 40]. AMIT has been demonstrated to decrease metabolic energy consumption and calf muscle activation [40, 41]. Drawing inspiration from AMIT, unilateral ankle torque assistance is applied to the musculoskeletal model in the simulation based on the characteristics of the biological ankle joint moment profile. The torque profile exhibits linear growth during the middle stance, follows a sine pattern during the terminal stance, and maintains zero during the early stance and swing phase [40]. This profile is modeled as

$$\tau_d = \begin{cases} 0, & 0 \leq T < T_i \\ \tau_m \cdot (T - T_i) / (T_m - T_i), & T_i \leq T < T_m \\ \tau_m + 0.5 \cdot p_{mp} \cdot (\tau_p - \tau_m), & T_m \leq T < T_p \\ 0.5 \cdot p_{pf} \cdot \tau_p, & T_p \leq T < T_f \\ 0, & T_f \leq T < 100 \end{cases} \quad (2)$$

$$\begin{cases} p_{mp} = 1 - \cos\left(\frac{\pi \cdot (T - T_p)}{T_p - T_m}\right) \\ p_{pf} = 1 + \cos\left(\frac{\pi \cdot (T - T_p)}{T_f}\right) \end{cases} \quad (3)$$

where T , T_i , T_m , and T_p represent the current gait phase, and the phases of assistance initiation, intermediate peak, and maximum peak, respectively (Fig. 2(a)). τ_i , τ_m , and τ_p denote the initial, intermediate peak, and maximum peak torques (Fig. 2(a)), while τ_d represents the current desired torque, p_{mp} and p_{pf} are the torque profiles in the ascending ($T_m \leq T < T_p$) and descending ($T_p \leq T < T_f$) stages of the terminal stance phase, respectively.

During optimization, T_i , T_m , and T_p serve as optimization parameters. These exoskeleton timing parameters are co-optimized with the neuromuscular controller parameters within the same CMA-ES loop, under the cost function J . They determine the temporal shape of the assistive torque profile and therefore indirectly affect the cost function by modulating the muscular effort required to maintain the prescribed walking task, which is reflected mainly in E_{CoT} and E_{muscle} . Through this joint optimization, the CMA-ES converges to a parameter set in which the neuromuscular system produces a stable and energy-efficient gait pattern that is well-matched to the assistive torque profile delivered by the exoskeleton. Given that the unilateral ankle exoskeleton used in this study weighs only 0.1 kg, which is negligible compared to the shank segment mass of 3.71 kg, its impact on gait dynamics is minimal. Therefore, in the simulation, rather than incorporating the physical exoskeleton or modifying the mass parameters of the assisted shank segment, the assistive plantarflexion is implemented by directly applying torque at the unilateral ankle joint to simulate an ideal massless exoskeleton. In addition, the human-exoskeleton interface is modeled as an

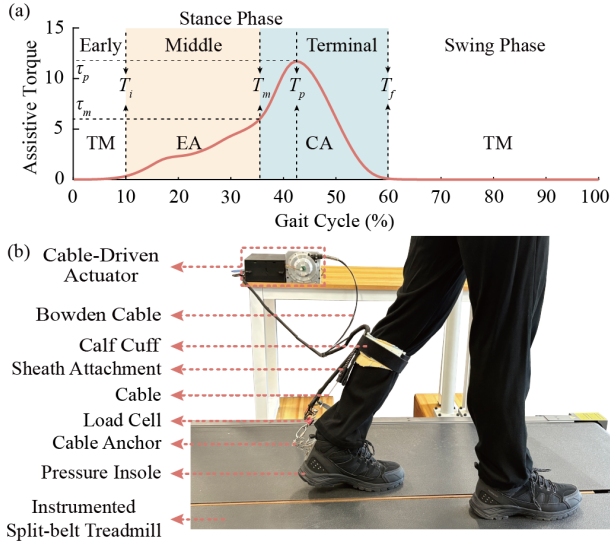


Fig. 2. Powered ankle exoskeleton platform. (a) Exoskeleton assistive torque profile. The exoskeleton provides eccentric assistance (EA) and concentric (CA) assistance during the middle and terminal stance phases, respectively. No assistance [transparent mode (TM)] is provided during the early stance and swing phases. (b) Leg-worn components and cable-driven actuator.

ideal rigid coupling at the ankle joint. The potential series compliance introduced by the cuff, soft tissue, or Bowden cable transmission is not explicitly modeled in the present predictive simulation framework.

In the analysis of assisted walking results, the biological ankle joint moment was calculated by subtracting the exoskeleton assistance torque from the total ankle joint moment [41, 42]. The muscle activations of the soleus (SOL) and gastrocnemius (GAS) were summed to represent the plantar flexor (PF) muscle group activation. The resulting PF muscle group envelope was normalized to the peak value observed for each participant during normal walking conditions. The muscle forces of SOL and GAS were summed to represent the PF muscle group force. Heel strike was identified based on GRF. The kinematics, kinetics, muscle activation, and muscle force were normalized to 0–100% of the gait cycle for each step. Mean and standard deviation values were then computed across multiple gait cycles.

III. EXPERIMENTAL VERIFICATION

A. Ankle Exoskeleton

As shown in Fig. 2(b), the ankle exoskeleton is designed to augment plantar flexor function during locomotion across different terrains. The system consists of leg-worn components including a calf cuff (0.11 kg), a custom shoe with a cable anchor (0.65 kg), and a pressure insole for foot-ground contact detection, as well as an off-board cable-driven actuator (1.10 kg) mounted on a stationary frame to minimize inertial influence on gait [40]. Mechanical power is transmitted through a Bowden cable, and an in-line load cell (FUTEK, USA) measures the actual cable force.

The actuator consists of a brushless DC motor integrated with a 6:1 planetary gearbox, offering a rated torque of 8 Nm, a peak torque of 16 Nm, and a rated speed of 300

r/min. The assistive force is delivered through a single-wrap pulley with a radius of 43.5 mm. A clutch-spring mechanism composed of a unidirectional mechanical clutch and a pre-tensioned torsion spring enables three operating modes: concentric assistance, eccentric assistance, and transparent mode. In concentric assistance, the motor spools the cable to deliver active plantarflexion torque. In eccentric assistance, the cable is passively unspooled and braking torque is generated through electromagnetic damping. In transparent mode, the clutch disengages and the ankle rotates freely with only the passive spring engaged. Motor and pulley positions are measured by magnetic encoders, and actuation is regulated by custom electronics integrating the motor driver and data acquisition board.

The exoskeleton adopts a three-level control architecture. The high-level controller performs gait phase estimation using a finite state machine based on plantar pressure signals sampled at 100 Hz for heel-strike detection, and the gait cycle duration is updated iteratively. The gait cycle is segmented into early stance, midstance, terminal stance, and swing phases. The mid-level controller generates phase-specific commands. During early stance, the clutch is engaged to prepare force transmission. During midstance, eccentric assistance is provided to resist excessive dorsiflexion, and the braking torque is modulated by the PWM duty cycle applied to the motor windings. During terminal stance, concentric assistance is delivered according to the desired plantarflexion torque profile τ_d defined in Eqs. (2) and (3). The torque profile is parameterized by the initial torque τ_i , intermediate peak torque τ_m , maximum peak torque τ_p , and their corresponding gait phases T_i , T_m , T_p , and T_f , as illustrated in Fig. 2(a). The low-level controller executes motor actuation through a PI current loop for clutch control, a PWM-based damping control scheme for eccentric assistance, and a PI cable-force closed-loop controller that regulates motor velocity to track the desired torque profile during terminal stance. The assistive torque delivered to the ankle joint is regulated in closed loop based on the in-line load cell feedback, thereby compensating for potential load-transfer losses caused by interface compliance and Bowden cable transmission. The torque definition and control parameters used in the experiments are consistent with those implemented in the predictive simulation to ensure comparability.

B. Experimental Protocol

To validate the effectiveness of the predictive simulation framework, we conducted experimental verification based on the simulation scenarios, including terrain, exoskeleton assistance, and walking speed. This investigation recruited 5 healthy male subjects [age(years)-height(m)-weight(kg): AB01: 26-1.78-72, AB02: 23-1.80-78, AB03: 23-1.81-78, AB04: 22-1.84-85, AB05: 22-1.78-101]. The same assistive torque profile was applied to all five participants. All participants were free from musculoskeletal injuries or conditions potentially affecting gait biomechanics. The experimental protocol was conducted on an instrumented treadmill at a constant velocity of 1.0 m/s, encompassing two primary experimental conditions: (1) normal walking without wearing the

exoskeleton (NoExo) and (2) assisted walking with a powered ankle exoskeleton (ExoOn). Each condition involved a 6-minute walking protocol, with the initial 3 minutes dedicated to participant familiarization and the subsequent 3 minutes allocated for comprehensive data acquisition. The 3-minute familiarization phase was designed to allow participants to adapt to the treadmill, the prescribed walking speed, the inclined surface, and the assistive torque profile, such that the subsequent 3-minute data-collection phase captured gait behavior after a stable adaptation state had been reached. All participants in this study were experienced subjects who had previously taken part in multiple exoskeleton-related experiments within our research group, and were therefore expected to converge to a stable gait pattern within a relatively short period. The duration of each phase was further determined as a compromise between ensuring sufficient familiarization, capturing a sufficient number of steady-state gait cycles, and minimizing fatigue effects, particularly under the energy-intensive inclined walking conditions. The experimental trials simulated three different terrain scenarios: level ground, 5-deg uphill, and 5-deg downhill walking. This research protocol received formal approval from the Local Ethics Committee following strict adherence to research ethics, and written informed consent was obtained from all subjects.

1) *Muscle Activation Evaluation*: Muscle activation was recorded via surface electromyography (EMG) electrodes (Delsys, USA) at 2 kHz sampling frequency. Measurements were obtained from four ankle-associated muscles on the assisted leg: tibialis anterior (TA), SOL, medial gastrocnemius (MG), and lateral gastrocnemius (LG). Electrode placement strictly adhered to SENIAM guidelines. Raw EMG signals underwent processing through a high-pass filter at 20 Hz, followed by full-wave rectification, and subsequent low-pass filtering at 6 Hz to produce linear envelopes that characterize muscle activity. The resulting EMG envelopes were normalized to the maximum activation level recorded for each individual muscle and subject during NoExo. The normalized MG and LG envelopes were summed to represent the GAS muscle group activation [43]. Similarly, the normalized SOL, MG, and LG signals were summed to represent the PF muscle group activation [43]. The resulting muscle group envelopes were subsequently normalized to their respective peak values observed during each subjects NoExo [43].

2) *Kinematics and Kinetics Assessment*: A motion capture system (VICON, UK) operating at 100 Hz was employed to acquire data using the sixteen-marker Plug-in-Gait protocol. An instrumented split-belt treadmill (Bertec Corporation, Ohio, USA) equipped with dual force plates provided ground reaction force measurements at 1000 Hz. Pre-processing and preparation of experimental data were conducted using the MOtoNMS toolbox within MATLAB R2022b (MathWorks, USA) [44]. The Gait2392 model in OpenSim (version 3.3) was personalized for each subject through anthropometric measurements obtained during a static motion capture trial [45]. Joint kinematics of the ankle joint of the assisted limb were computed through inverse kinematics using the captured marker trajectories. The integration of these kinematic data with ground reaction forces enabled joint moment calculations via

inverse dynamics. Muscle-tendon unit (MTU) kinematics were derived through subsequent muscle analysis. For the assisted limb, the biological ankle joint moment was calculated by subtracting the measured exoskeleton assistive torque (obtained from the load cell) from the total ankle joint moment [42].

3) *Muscle Force Estimation*: The CEINMS facilitated the estimation of muscle forces acting on the assisted limb's ankle joint [46]. The CEINMS model was calibrated using three walking trials isolated from each participants walking experiments under different conditions. These calibration trials were excluded from the final analysis to avoid potential bias in the results [46, 47]. Following calibration, the CEINMS operated in EMG-assisted mode, optimizing the balance between matching joint moment derived from inverse dynamics and incorporating experimental muscle excitation measurements:

$$F_{obj} = \alpha \cdot \sum_{k \in DOFs} (\tau_k - \tilde{\tau}_k)^2 + \beta \cdot \sum_{j \in MTUs} (e_j - \tilde{e}_j)^2, \quad (4)$$

where $(\tau_k - \tilde{\tau}_k)^2$ represents the squared difference between estimated and experimental biological ankle moment, while $(e_j - \tilde{e}_j)^2$ quantifies the squared deviation between synthesized and experimental muscle activations. The weighting coefficients (α, β) are used to determine the relative weights for minimizing the error terms.

The CEINMS was driven in EMG-assisted mode using EMG signals from TA, SOL, MG, and LG, as well as the MTU kinematics and joint moment. This operational mode performs minimal adjustments to the recorded EMG. Within the CEINMS framework, muscle forces were computed using a Hill-type muscle model. These coefficients (α, β) were set based on previous research to minimize the root mean square error between predicted and experimental results $(\alpha = 10,000, \beta = 10)$ [48, 49]. The GAS muscle group force was calculated by summing the muscle forces of MG and LG. The PF muscle group force was calculated by summing the muscle forces of MG, LG, and SOL.

C. Data Analysis

Walking experiment data were segmented into individual gait cycles based on ground reaction force measurements, and each cycle was subsequently time normalized to 0 to 100 percent of the gait cycle. For each subject and each walking condition, the first 20 consecutive strides that satisfied the criteria for stable and precise segmentation were retained [43], and a subject specific and condition specific mean waveform was computed across these 20 strides. The same segmentation and normalization procedure were applied to the simulation outputs, where each 20 second continuous gait sequence was divided into individual gait cycles to obtain a subject specific mean waveform for every condition. The group level results were obtained by averaging the subject specific mean waveforms across the five participants, with the shaded areas representing the inter subject standard deviation.

The effectiveness of the framework was quantitatively evaluated across four dimensions. First, the maximum absolute normalized cross-correlation (MANCC) coefficient was calculated to assess temporal similarity between experimental

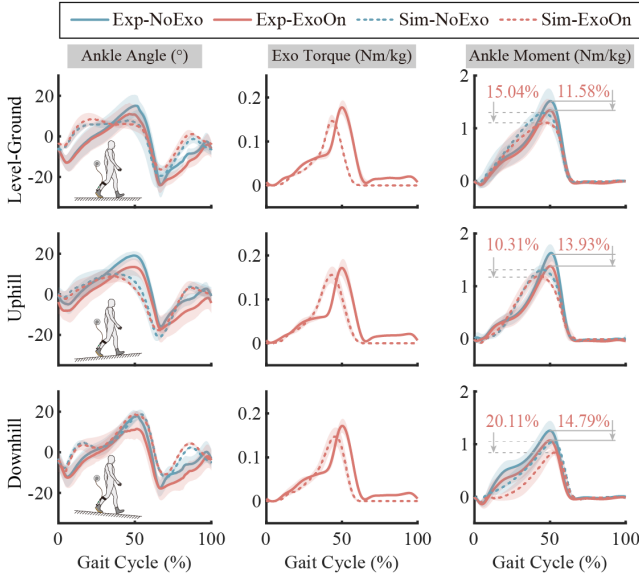


Fig. 3. Kinematic and kinetic results during level-ground (top), uphill (middle), and downhill (bottom) walking. The lines in each subplot represent the average results under four different conditions: normal walking without exoskeleton under experimental conditions (Exp-NoExo, blue solid line), assisted walking with powered exoskeleton under experimental conditions (Exp-ExoOn, red solid line), normal walking without exoskeleton under simulated conditions (Sim-NoExo, blue dashed line), and assisted walking with powered exoskeleton under simulated conditions (Sim-ExoOn, red dashed line). The shaded areas represent the standard deviation (SD). The experimental data are averaged across five subjects.

and simulation results, including kinematics, kinetics, muscle activation, and muscle force during NoExo and ExoOn across three terrains, as well as exoskeleton assistive moment (M_{exo}) during ExoOn. MANCC values exceeding 0.90, 0.70, and 0.40 indicated excellent, strong, and moderate similarity, respectively [50, 51]. Second, the Pearson correlation coefficient (PCC) was computed to determine the correlation between experimental and simulation results, with ranges consistent with those assessed by MANCC [31, 52]. Third, the relative root mean squared error (rRMSE) between experimental and simulation results was calculated to evaluate similarity across kinematics, kinetics, and PF muscle force [17]. Finally, the percentage changes in peak and mean values for kinetics, PF muscle activation, and PF muscle forces from NoExo to ExoOn were computed for both experimental and simulation results to assess similarity [48, 53]. Statistical significance between different conditions was evaluated using paired t-tests, with the significance level set at $\alpha = 0.05$ for all analyses.

IV. RESULTS

A. Kinematics and Kinetics

Under NoExo, the characteristic plantarflexion valleys of ankle joint angles occurred at 67.1% and 64.3% of the gait cycle for level-ground walking, 67.1% and 64.1% for uphill walking, and 66.5% and 69.1% for downhill walking in experimental and simulation results, respectively (Fig. 3). Under ExoOn, the plantarflexion valleys occurred at 66.7% and 65.3% of the gait cycle for level-ground walking, 67.1% and 62.5% for uphill walking, and 67.1% and 67.8% for

TABLE I
ASSISTIVE TORQUE PARAMETERS ACROSS SUBJECTS AND TERRAINS

Subjects	Level-Ground		Uphill		Downhill	
	τ_p	τ_m	τ_p	τ_m	τ_p	τ_m
AB01	0.20	0.10	0.20	0.07	0.19	0.09
AB02	0.19	0.08	0.19	0.08	0.20	0.08
AB03	0.18	0.08	0.18	0.08	0.18	0.08
AB04	0.19	0.09	0.19	0.08	0.18	0.09
AB05	0.16	0.07	0.15	0.08	0.15	0.08
Mean	0.18	0.08	0.18	0.08	0.18	0.08
SD	0.02	0.01	0.02	0.00	0.02	0.01

τ_p : maximum peak torque, τ_m : intermediate peak torque, SD: standard deviation; Unit: Nm/kg.

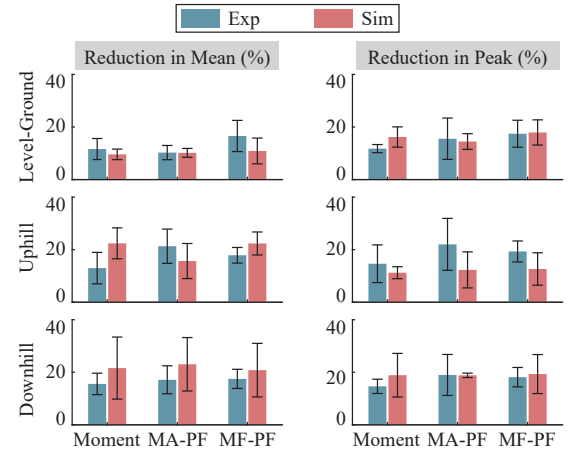


Fig. 4. Percentage changes in mean and peak values of joint moment, PF muscle activation (MA-PF), and PF muscle force (MF-PF) due to exoskeleton assistance during level-ground, uphill, and downhill walking between experimental and simulation results. Each group of bars represents results under two different conditions: experimental (Exp, $N = 5$, Mean $\pm$ SD) and simulation (Sim, $N = 5$, Mean $\pm$ SD). No significant differences were observed in the mean and peak value changes between experimental and simulation results.

downhill walking. With exoskeleton assistance, the peak of ankle joint moment decreased by 11.58% and 15.04% (level-ground), 13.93% and 10.31% (uphill), and 14.79% and 20.11% (downhill) in experimental and simulation results, respectively (Fig. 3). Table I summarizes the assistive torque parameters for each participant across all tested terrains.

As shown in Table II, MANCC values for ankle joint angles exceeded 0.70 in all conditions except downhill walking with ExoOn (0.66). Ankle joint moment MANCC values exceeded 0.90 across all conditions. MANCC values for M_{exo} exceeded 0.95 in all conditions. As shown in Table III, PCC values for ankle joint angles exceeded 0.60 in all conditions except uphill walking with ExoOn (0.43). PCC values for ankle joint moment exceeded 0.83 across all conditions. PCC values for M_{exo} exceeded 0.50 in all conditions.

As shown in Table IV, the rRMSE values for ankle joint angles under NoExo and ExoOn were 0.25 and 0.29 for level-ground walking ($p = 0.494$), 0.27 and 0.32 for uphill walking ($p = 0.234$), and 0.17 and 0.29 for downhill walking ($p = 0.039$). The rRMSE values for ankle joint moment were 0.19 and 0.17 for level-ground walking ($p = 0.724$), 0.15 and 0.20

TABLE II
THE MANCC BETWEEN EXPERIMENTAL AND SIMULATION RESULTS

Item	Level-Ground		Uphill		Downhill	
	NoExo	ExoOn	NoExo	ExoOn	NoExo	ExoOn
Ankle Angle	0.79 ± 0.16	0.72 ± 0.11	0.76 ± 0.07	0.84 ± 0.07	0.78 ± 0.07	0.66 ± 0.10
Ankle Moment	0.98 ± 0.01	0.96 ± 0.03	0.97 ± 0.00	0.97 ± 0.01	0.97 ± 0.01	0.96 ± 0.02
MA-TA	0.79 ± 0.07	0.82 ± 0.09	0.82 ± 0.07	0.85 ± 0.07	0.84 ± 0.07	0.86 ± 0.07
MA-GAS	0.89 ± 0.07	0.91 ± 0.05	0.89 ± 0.02	0.88 ± 0.02	0.93 ± 0.05	0.87 ± 0.09
MA-SOL	0.94 ± 0.03	0.91 ± 0.06	0.94 ± 0.03	0.95 ± 0.02	0.91 ± 0.06	0.86 ± 0.05
MA-PF	0.93 ± 0.04	0.93 ± 0.04	0.92 ± 0.02	0.92 ± 0.02	0.94 ± 0.02	0.88 ± 0.07
MF-TA	0.85 ± 0.05	0.88 ± 0.08	0.91 ± 0.03	0.94 ± 0.01	0.85 ± 0.03	0.90 ± 0.03
MF-GAS	0.95 ± 0.02	0.96 ± 0.03	0.94 ± 0.03	0.94 ± 0.03	0.97 ± 0.01	0.89 ± 0.07
MF-SOL	0.97 ± 0.00	0.93 ± 0.05	0.94 ± 0.02	0.95 ± 0.02	0.95 ± 0.04	0.95 ± 0.02
MF-PF	0.98 ± 0.01	0.96 ± 0.03	0.97 ± 0.01	0.96 ± 0.01	0.98 ± 0.02	0.96 ± 0.02
M_{exo}	–	0.97 ± 0.00	–	0.96 ± 0.01	–	0.97 ± 0.01

MA: muscle activation, MF: muscle force, M_{exo} : exoskeleton assistive torque, MANCC: maximum absolute normalized cross-correlation.

TABLE III
THE PCC BETWEEN EXPERIMENTAL AND SIMULATION RESULTS

Item	Level-Ground		Uphill		Downhill	
	NoExo	ExoOn	NoExo	ExoOn	NoExo	ExoOn
Ankle Angle	0.66 ± 0.31	0.66 ± 0.18	0.60 ± 0.15	0.43 ± 0.19	0.82 ± 0.06	0.77 ± 0.08
Ankle Moment	0.85 ± 0.10	0.89 ± 0.09	0.88 ± 0.06	0.83 ± 0.07	0.94 ± 0.04	0.91 ± 0.03
MA-TA	0.52 ± 0.13	0.57 ± 0.19	0.54 ± 0.13	0.59 ± 0.17	0.63 ± 0.13	0.65 ± 0.15
MA-GAS	0.70 ± 0.22	0.64 ± 0.17	0.77 ± 0.11	0.77 ± 0.10	0.40 ± 0.11	0.38 ± 0.17
MA-SOL	0.80 ± 0.03	0.69 ± 0.12	0.85 ± 0.06	0.85 ± 0.09	0.50 ± 0.23	0.50 ± 0.16
MA-PF	0.78 ± 0.10	0.71 ± 0.09	0.83 ± 0.08	0.81 ± 0.10	0.48 ± 0.18	0.48 ± 0.18
MF-TA	0.55 ± 0.16	0.68 ± 0.23	0.73 ± 0.13	0.79 ± 0.05	0.61 ± 0.09	0.70 ± 0.07
MF-GAS	0.83 ± 0.11	0.89 ± 0.10	0.81 ± 0.10	0.75 ± 0.10	0.86 ± 0.06	0.55 ± 0.26
MF-SOL	0.83 ± 0.15	0.83 ± 0.12	0.82 ± 0.08	0.76 ± 0.10	0.88 ± 0.14	0.84 ± 0.07
MF-PF	0.84 ± 0.13	0.87 ± 0.09	0.86 ± 0.07	0.78 ± 0.10	0.91 ± 0.10	0.84 ± 0.08
M_{exo}	–	0.57 ± 0.03	–	0.54 ± 0.14	–	0.72 ± 0.07

MA: muscle activation, MF: muscle force, M_{exo} : exoskeleton assistive torque, PCC: Pearson correlation coefficient.

TABLE IV
THE RELATIVE RMSE BETWEEN EXPERIMENTAL AND SIMULATION RESULTS

Item	Level-Ground			Uphill			Downhill		
	NoExo	ExoOn	<i>p</i> -value	NoExo	ExoOn	<i>p</i> -value	NoExo	ExoOn	<i>p</i> -value
Ankle Angle	0.25 ± 0.12	0.29 ± 0.08	0.494	0.27 ± 0.07	0.32 ± 0.06	0.234	0.17 ± 0.04	0.29 ± 0.11	0.039
Ankle Moment	0.19 ± 0.06	0.17 ± 0.07	0.724	0.15 ± 0.03	0.20 ± 0.05	0.075	0.13 ± 0.05	0.16 ± 0.05	0.524
MF-PF	0.21 ± 0.08	0.22 ± 0.12	0.848	0.17 ± 0.04	0.23 ± 0.08	0.134	0.14 ± 0.05	0.19 ± 0.05	0.059

MF: muscle force.

for uphill walking ($p = 0.075$), and 0.13 and 0.16 for downhill walking ($p = 0.524$).

As shown in Fig. 4, the changes in mean ankle joint moment due to exoskeleton assistance were $11.63 \pm 4.00\%$ and $9.59 \pm 2.01\%$ for level-ground walking ($p = 0.156$), $12.97 \pm 5.98\%$ and $22.40 \pm 5.89\%$ for uphill walking ($p = 0.091$), and $15.57 \pm 4.08\%$ and $21.59 \pm 11.81\%$ for downhill walking ($p = 0.316$) in experimental and simulation results, respectively. The changes in peak ankle joint moment were $11.78 \pm 1.55\%$ and $16.20 \pm 3.86\%$ for level-ground walking ($p = 0.086$),

$14.64 \pm 7.17\%$ and $11.22 \pm 2.28\%$ for uphill walking ($p = 0.429$), and $14.64 \pm 2.71\%$ and $18.88 \pm 8.30\%$ for downhill walking ($p = 0.275$).

B. Muscle Activation

Under NoExo, the characteristic peaks of PF muscle activations occurred at 45.9% and 47.1% of the gait cycle for level-ground walking, 46.5% and 46.6% for uphill walking, and 46.2% and 55.2% for downhill walking in experimental and simulation results, respectively (Fig. 5). Under ExoOn,

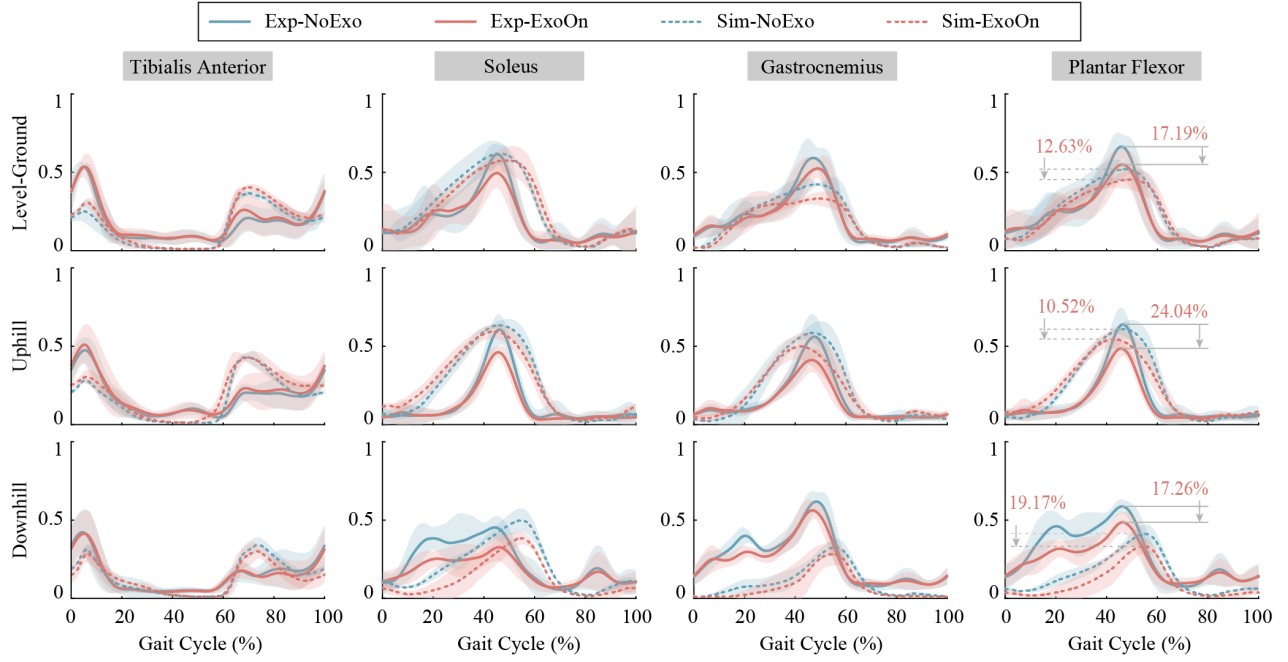


Fig. 5. Muscle activations of the tibialis anterior, soleus, gastrocnemius, and plantar flexor during level-ground, uphill, and downhill walking. The lines in each subplot represent the average results under four different conditions: Exp-NoExo, Exp-ExoOn, Sim-NoExo, Sim-ExoOn. The shaded areas represent the standard deviation (SD). The experimental data are averaged across five subjects.

the peaks occurred at 46.1% and 49.2% of the gait cycle for level-ground walking, 45.8% and 43.1% for uphill walking, and 46.4% and 54.9% for downhill walking. With exoskeleton assistance, the peak of the PF muscle activation decreased by 17.19% and 12.63% (level-ground), 24.04% and 10.52% (uphill), and 17.26% and 19.17% (downhill) in experimental and simulation results, respectively (Fig. 5).

As shown in Table II, MANCC values for MA-TA exceeded 0.80 in all conditions except level-ground walking with NoExo (0.79). MANCC values for MA-GAS exceeded 0.80 across all conditions, with level-ground walking with ExoOn (0.91) and downhill walking with NoExo (0.93) exceeding 0.90. MANCC values for MA-SOL exceeded 0.90 in all conditions except downhill walking with ExoOn (0.86). MANCC values for MA-PF exceeded 0.90 in all conditions except downhill walking with ExoOn (0.88).

As shown in Table III, PCC values for MA-TA exceeded 0.52 across all conditions, with downhill walking (NoExo: 0.63, ExoOn: 0.65) surpassing 0.60. PCC values for MA-GAS exceeded 0.60 in all conditions except downhill walking (NoExo: 0.40, ExoOn: 0.38), reaching 0.77 during uphill walking. PCC values for MA-SOL exceeded 0.60 in all conditions except downhill walking (0.50), with uphill walking achieving 0.85. PCC values for MA-PF exceeded 0.70 in all conditions except downhill walking (0.48).

As shown in Fig. 4, the changes in mean muscle activation of PF due to exoskeleton assistance were $10.27 \pm 2.71\%$ and $10.17 \pm 1.68\%$ for level-ground walking ($p = 0.954$), $21.30 \pm 6.53\%$ and $15.67 \pm 6.67\%$ for uphill walking ($p = 0.242$), and $17.13 \pm 5.33\%$ and $23.04 \pm 10.17\%$ for downhill walking ($p = 0.449$) in experimental and simulation results, respectively. The changes in peak muscle activation of PF were

$15.54 \pm 7.87\%$ and $14.48 \pm 2.99\%$ for level-ground walking ($p = 0.850$), $22.01 \pm 9.89\%$ and $12.30 \pm 6.86\%$ for uphill walking ($p = 0.106$), and $18.98 \pm 7.80\%$ and $18.83 \pm 0.84\%$ for downhill walking ($p = 0.971$).

C. Muscle Forces

Under NoExo, the characteristic peaks of PF muscle forces occurred at 49.4% and 43.9% of the gait cycle for level-ground walking, 50.0% and 43.1% for uphill walking, and 49.2% and 51.1% for downhill walking in experimental and simulation results, respectively (Fig. 6). Under ExoOn, the peaks occurred at 48.5% and 44.7% of the gait cycle for level-ground walking, 49.4% and 39.0% for uphill walking, and 49.3% and 51.1% for downhill walking. With exoskeleton assistance, the peak of the PF muscle forces decreased by 19.48% and 16.74% (level-ground), 19.07% and 10.96% (uphill), and 18.89% and 20.71% (downhill) in experimental and simulation results, respectively (Fig. 6).

As shown in Table II, MANCC values for MF-TA exceeded 0.80 across all conditions. MANCC values for MF-GAS exceeded 0.90 in all conditions except downhill walking with ExoOn (0.89). MANCC values for MF-SOL and MF-PF exceeded 0.90 across all conditions. As shown in Table III, PCC values for MF-TA exceeded 0.60 in all conditions except level-ground walking with NoExo (0.55), with uphill walking conditions (NoExo: 0.73, ExoOn: 0.79) surpassing 0.70. PCC values for MF-GAS exceeded 0.80 in all conditions except uphill walking with ExoOn (0.75) and downhill walking with ExoOn (0.55). PCC values for MF-SOL exceeded 0.80 in all conditions except uphill walking with ExoOn (0.76). PCC values for MF-PF exceeded 0.80 in all conditions except uphill walking with ExoOn (0.78), with downhill walking with

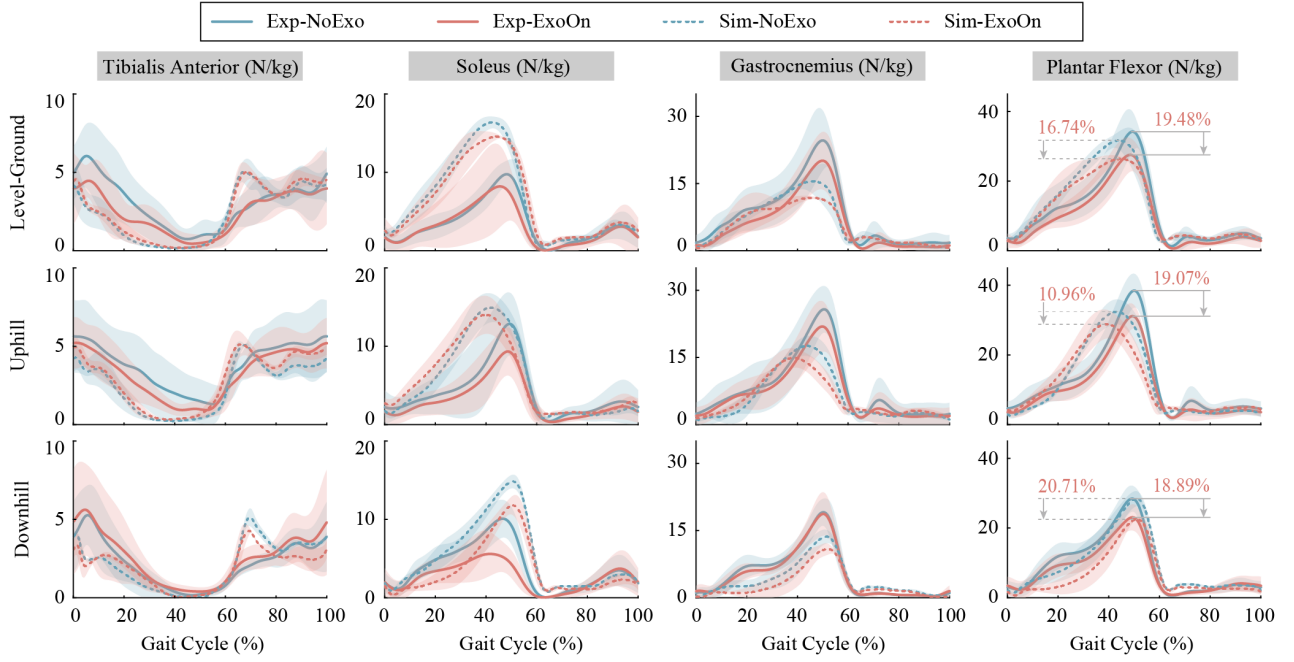


Fig. 6. Muscle forces of the tibialis anterior, soleus, gastrocnemius, and plantar flexor during level-ground, uphill, and downhill walking. The lines in each subplot represent the average results under four different conditions: Exp-NoExo, Exp-ExoOn, Sim-NoExo, Sim-ExoOn. The shaded areas represent the standard deviation (SD). The experimental data are averaged across five subjects.

NoExo (0.91) exceeding 0.90. As shown in Table IV, The rRMSE values for MF-PF under NoExo and ExoOn were 0.21 and 0.22 for level-ground walking ($p = 0.848$), 0.17 and 0.23 for uphill walking ($p = 0.134$), and 0.14 and 0.19 for downhill walking ($p = 0.059$).

As shown in Fig. 4, the changes in mean muscle forces of PF due to exoskeleton assistance were $16.58 \pm 5.94\%$ and $10.89 \pm 4.90\%$ for level-ground walking ($p = 0.094$), $17.85 \pm 2.97\%$ and $22.35 \pm 4.38\%$ for uphill walking ($p = 0.126$), and $17.48 \pm 3.65\%$ and $20.81 \pm 10.18\%$ for downhill walking ($p = 0.518$) in experimental and simulation results, respectively. The changes in peak muscle forces of PF were $17.45 \pm 5.13\%$ and $17.93 \pm 4.80\%$ for level-ground walking ($p = 0.926$), $19.31 \pm 3.99\%$ and $12.63 \pm 6.16\%$ for uphill walking ($p = 0.051$), and $18.13 \pm 3.69\%$ and $19.31 \pm 7.43\%$ for downhill walking ($p = 0.739$).

V. DISCUSSION

This study presents a predictive simulation framework for evaluating the biomechanical effects of exoskeleton-assisted walking, particularly focusing on muscle activation and muscle force analysis. The predictions of changes in joint kinetics, PF muscle activation, and PF muscle force due to exoskeleton assistance across different terrains were generally consistent with experimental results.

A. Kinematic and Kinetic Changes

In terms of kinematics, the phase difference of ankle joint plantarflexion valley values between experimental and simulation results ranged from 0.7% to 4.6%, indicating that the simulated ankle joint kinematic profiles generally matched

the temporal characteristics of plantarflexion valley observed in experimental results (Fig. 3). Across the entire kinematic profile, the simulation demonstrated strong temporal similarity with experimental results (most MANCC > 0.70) and good correlation (most PCC > 0.60) [31, 50–52]. The rRMSE of joint angles under ExoOn was greater than that under NoExo, with significant differences observed only during downhill walking ($p = 0.039$). This discrepancy may be attributed to the increased complexity of constraints during downhill walking, which likely involves multi-objective conflicting requirements such as balancing gravitational potential energy utilization with motor control. However, the simplified reflex-based control model employed in this study may have been insufficient to fully capture these complex constraints. Consequently, the simulation predicted minimal kinematic changes between ExoOn and NoExo, whereas substantial differences were observed experimentally during the assistance phase, including a reduction in dorsiflexion peak values [40, 41]. This kinematic discrepancy between simulation and experiment is further reflected in the torque profiles. As shown in Table I and Fig. 3, the experimental peak assistive torques showed little variation among participants and terrains, yet they were uniformly larger in magnitude and delayed in timing compared with the simulated profiles. Since the preset torque commands were identical in both conditions, these differences are most likely attributable to the aforementioned kinematic discrepancies (e.g., joint angle trajectories), which may have altered the effective load transmission or the actual torque delivered through the exoskeleton. Collectively, this limitation potentially contributed to the reduced prediction accuracy under ExoOn.

In terms of kinetics, the simulation demonstrated excellent

temporal similarity with experimental results (MANCC > 0.90) and good correlation (PCC > 0.83) [31, 50–52]. The rRMSE of joint moment under ExoOn was lower than that under NoExo during level-ground walking, whereas it was higher during both uphill and downhill walking. However, no significant differences in joint moment rRMSE between NoExo and ExoOn were observed across the three terrains ($p > 0.05$), indicating that exoskeleton assistance did not significantly affect simulation accuracy.

Despite the overall consistency in temporal characteristics and the general trends of joint angle and joint moment waveforms shown in Fig. 3, visible differences in curve shapes between experimental and simulation results can still be observed. These shape differences mainly arise from three aspects of the proposed framework. First, the adopted control policy optimization based on a reflex-based neuromuscular controller constrains the space of admissible control inputs, and therefore tends to produce smoother and more stereotyped waveforms than those obtained from data-tracking trajectory optimization methods, as also reflected in the comparison reported in Table V. Second, the H0918 two-dimensional musculoskeletal model restricts motion to the sagittal plane and represents foot-ground interaction with two visco-elastic Hunt-Crossley contact spheres, which simplifies the actual three-dimensional kinematics and the distributed plantar pressure observed in experiments, leading to local deviations especially around heel-strike and push-off events. Third, each simulated waveform is generated from a single optimized control policy under a given condition, whereas the experimental waveforms are averaged across five subjects and inherently contain inter-subject and intra-subject variability that cannot be fully reproduced by a single deterministic simulation. These factors jointly contribute to the residual shape differences observed in Fig. 3, while the consistency of the key biomechanical metrics evaluated in this study (MANCC, PCC, rRMSE, and the percentage changes in mean and peak values) is preserved.

As shown in Fig. 4, the changes in mean and peak values of joint moment remained at comparable levels between experimental and simulation results. For mean values, the magnitude of changes during uphill (9.43%) and downhill (6.02%) walking was greater than that during level-ground walking (2.04%). For peak values, the magnitude of changes was similar across level-ground (4.42%), uphill (3.42%), and downhill (4.24%) walking conditions. Neither mean nor peak value changes showed significant differences between simulation and experimental results ($p > 0.05$). These findings suggest that the simulation could reasonably predict the trend of joint moment reduction induced by exoskeleton assistance, although incline walking may potentially affect the prediction accuracy of joint moment mean value changes.

The consistency between the measured and simulated exoskeleton assistive torques (M_{exo}) was also examined to verify that the assistance applied in the experiments faithfully reproduced the torque profile prescribed in the simulation. The MANCC values exceeded 0.95 across all three terrains, indicating excellent temporal similarity, while the PCC values ranged from 0.54 to 0.72, reflecting moderate to strong amplitude correlations. The slightly lower PCC values can be

attributed to small fluctuations in the measured cable force introduced by Bowden cable friction and the closed-loop tracking error of the force controller. These results confirm that the experimental device reproduces the desired torque profile with high fidelity, ensuring that the assistive input applied in the experiments closely matches the one prescribed in the simulation framework.

B. Muscle Activation Changes

In terms of muscle activation, TA and GAS demonstrated strong temporal similarity between experimental and simulation results (MANCC > 0.70) and good correlation (most PCC > 0.40), while SOL and PF exhibited excellent temporal similarity (most MANCC > 0.90) and good correlation (most PCC > 0.70) [31, 50–52].

As with the joint kinematics and kinetics shown in Fig. 3, visible curve shape differences between experimental and simulation muscle activation profiles can also be observed in Fig. 5. In addition to the general factors discussed in the Kinematic and Kinetic Changes subsection, two muscle-activation-specific factors further contribute to these differences. First, the reflex-based controller generates muscle excitations from a parsimonious set of length-, velocity-, and force-related feedback laws, which inherently produces smoother and more stereotyped excitation envelopes than the experimental EMG envelopes, the latter being affected by physiological noise, motor unit recruitment variability, and electrode-related effects such as crosstalk between adjacent muscles. Second, the activation dynamics within the Hill-type muscle model represent a lumped approximation of the underlying excitation-to-activation transformation, which does not fully capture the temporal richness of the recorded EMG. Despite these shape differences, the high MANCC and PCC values reported above indicate that the simulation reasonably captures the principal temporal and amplitude features of plantar flexor muscle activation under exoskeleton assistance.

As shown in Fig. 4, the changes in mean and peak values of PF muscle activation remained at comparable levels between experimental and simulation results. For mean values, the magnitude of changes during uphill (5.63%) and downhill (5.91%) walking was greater than that during level-ground walking (0.10%). For peak values, the magnitude of change during uphill walking (9.71%) was greater than that during level-ground (1.06%) and downhill (0.15%) walking. Neither mean nor peak value changes showed significant differences between simulation and experimental results ($p > 0.05$). These findings suggest that the simulation could reasonably predict the trend of PF muscle activation reduction induced by exoskeleton assistance, although uphill walking may potentially affect the prediction accuracy of PF muscle activation mean value changes, and incline walking may influence the prediction accuracy of PF muscle activation peak value changes.

C. Muscle Force Changes

In terms of muscle force, TA demonstrated strong temporal similarity between experimental and simulation results (MANCC > 0.70) and good correlation (most PCC > 0.60),

TABLE V
COMPARISON WITH RELATED STUDIES ON PREDICTIVE EFFECTS OF KINEMATICS, KINETICS, MUSCLE ACTIVATION, AND MUSCLE FORCE

Method	Study	Joint Angle	Joint Moment	Muscle Activation	Muscle Force
TO	[54]	–	–	–	–
	[20]	Baseline: – Adaptation: #	–	–	–
	[55]	Baseline: #, * (RMSE = 6.2) Adaptation: #	Baseline: #, * (RMSE = 27.1 Nm) Adaptation: –	–	–
	[56]	Baseline: #, * (RMSE = 2.64) Adaptation: #, * (RMSE = 8.28)	Baseline: – Adaptation: #	Baseline: – Adaptation: #	–
DTTO	[19]	Baseline: # Adaptation: #	–	Baseline: – Adaptation: #	–
	[17]	Baseline: #, * (rRMSE = 0.16) Adaptation: #, * (rRMSE = 0.18)	Baseline: #, * (rRMSE = 0.12) Adaptation: #, * (rRMSE = 0.20)	–	–
CPO	[31]	Baseline: – Adaptation: #, * PCC(L) = 0.6, PCC(U) = 0.43	–	Baseline: – Adaptation: #, * 0.01 ≤ PCC(L) ≤ 0.94, 0.11 ≤ PCC(U) ≤ 0.84	–
	This study	Baseline: #, * PCC(L) = 0.66, rRMSE(L) = 0.25; PCC(U) = 0.60, rRMSE(U) = 0.27 Adaptation: #, * PCC(L) = 0.66, rRMSE(L) = 0.29; PCC(U) = 0.43, rRMSE(U) = 0.32	Baseline: #, * rRMSE(L) = 0.19 Adaptation: #, * rRMSE(L) = 0.17	Baseline: #, * 0.52 ≤ PCC(L) ≤ 0.80, 0.54 ≤ PCC(U) ≤ 0.85 Adaptation: #, * 0.57 ≤ PCC(L) ≤ 0.69, 0.59 ≤ PCC(U) ≤ 0.85	Baseline: #, * Adaptation: #, * (L/U/D: Percentage changes in mean and peak values of PF ($p > 0.05$))

TO: trajectory optimization, DTTO: data-tracking trajectory optimization, CPO: control policy optimization. #: qualitative, *: quantitative.
L: level-ground walking, U: uphill walking. D: downhill walking. If there is no special mark, it means level-ground walking.

while GAS, SOL, and PF exhibited excellent temporal similarity (GAS: most MANCC > 0.90, SOL and PF: MANCC > 0.90) and good correlation (most PCC > 0.80) [31, 50–52]. The rRMSE of PF muscle force under ExoOn was greater than that under NoExo, although no significant differences were observed ($p > 0.05$), indicating that exoskeleton assistance did not significantly affect simulation accuracy.

Analogous to the shape differences observed in joint kinematics, kinetics (Fig. 3), and muscle activation (Fig. 5), visible discrepancies between experimental and simulation muscle force profiles can also be seen in Fig. 6. Beyond the factors discussed in the previous subsections, two muscle-force-specific factors further contribute to these differences. First, the simulation muscle forces are obtained directly from the forward dynamic simulation driven by the reflex-based neuromuscular controller, whereas the experimental muscle forces are estimated through CEINMS in EMG-assisted mode using subject-specific calibration. The two pipelines rely on different modeling assumptions, particularly the trade-off between matching the inverse-dynamics joint moment and respecting the recorded EMG, which is governed by the weighting coefficients (α , β). Second, deviations between the generic muscle parameters of the H0918 model (e.g., physiological cross-sectional area, optimal fiber length, tendon slack length,

and moment arms) and subject-specific anatomy may further influence the predicted force magnitudes and waveform shapes. Despite these shape differences, the high MANCC values (most > 0.90), the strong PCC values (most > 0.80), and the comparable rRMSE values between NoExo and ExoOn ($p > 0.05$) indicate that the framework reliably captures the principal temporal and amplitude features of plantar flexor muscle force under exoskeleton assistance, which is a central focus of this muscle-effort-oriented study.

As shown in Fig. 4, the changes in mean and peak values of PF muscle force remained at comparable levels between experimental and simulation results. For mean values, the magnitude of changes was similar across level-ground (5.69%), uphill (4.50%), and downhill (3.33%) walking conditions. For peak values, the magnitude of change during uphill walking (6.68%) was greater than that during level-ground (0.45%) and downhill (1.18%) walking. Neither mean nor peak value changes showed significant differences between experimental and simulation results ($p > 0.05$). These findings suggest that the simulation could reasonably predict the trend of PF muscle force reduction induced by exoskeleton assistance, although uphill walking may potentially affect the prediction accuracy of PF muscle force peak value changes.

D. Comparison with Related Studies

Predictive simulation has made significant progress in normal walking, accurately reproducing kinematics, kinetics, and metabolic energy expenditure [29]. However, studies on the effects of assistive devices such as exoskeletons on walking remain relatively limited [55]. As shown in Table V, existing predictive simulation methods can generally reproduce the key characteristics of kinematics, kinetics, and muscle activation during both normal walking and exoskeleton-assisted walking [14, 19, 31]. The fidelity of simulated motion is typically assessed by comparing simulated and experimental kinematics, kinetics, and muscle activation [15]. However, such validation work is often qualitative and typically focuses on level-ground walking conditions [15, 19, 20]. This study provides both qualitative and quantitative analyses through corresponding experimental validation to determine whether predictive simulation can replicate the biomechanical changes induced by exoskeleton assistance during multi-terrain walking.

Studies based on data-tracking trajectory optimization (DTTO) method have demonstrated notable effectiveness in replicating experimental kinematic and kinetic data during normal walking [17]. As shown in Table V, the similarity between experimental and simulation kinematic results in this study [Baseline: $rRMSE(L) = 0.25$; Adaptation: $PCC(L) = 0.66$, $rRMSE(L) = 0.29$, $PCC(U) = 0.43$] was comparable to other studies based on control policy optimization (CPO) method [Adaptation: $PCC(L) = 0.6$, $PCC(U) = 0.43$], though lower than DTTO-based methods (Baseline: $rRMSE = 0.16$; Adaptation: $rRMSE = 0.18$) [17, 31]. The similarity between experimental and simulation kinetics results during normal walking (Baseline: $rRMSE = 0.19$) was lower than DTTO-based methods (Baseline: $rRMSE = 0.12$), while showing better performance during exoskeleton-assisted walking (Adaptation: $rRMSE = 0.17$) compared to DTTO-based methods (Adaptation: $rRMSE = 0.20$) [17]. The lower similarity in kinematics and kinetics, compared to DTTO-based predictive simulations, can likely be attributed to the CPO-based predictive simulation constraining the space of possible inputs (Table V of [15]).

As shown in Table V, the similarity between experimental and simulation muscle activation results during exoskeleton-assisted walking [Adaptation: $0.59 \leq PCC(U) \leq 0.85$] was comparable to other CPO-based methods [Adaptation: $0.11 \leq PCC(U) \leq 0.84$] [31]. In contrast to previous predictive simulations, which have rarely addressed muscle force analysis, this study additionally evaluates PF muscle force adaptations and reasonably predicts the trend of PF muscle force reduction induced by exoskeleton assistance across different terrains (percentage changes $p > 0.05$). Overall, by jointly considering NoExo and ExoOn conditions across level-ground, uphill, and downhill walking, and by validating kinematics, kinetics, muscle activation, and muscle force through both qualitative and quantitative analyses, this study extends the scope of existing exoskeleton-related predictive simulation research.

E. Limitations and Future Work

While the proposed framework demonstrates promising predictive capabilities for the trends of biomechanical changes during exoskeleton-assisted walking across multiple terrains, several limitations should be acknowledged. First, direct experimental validation of muscle forces via non-invasive means is inherently difficult, and thus the present study relies on CEINMS-estimated forces for comparison. Although this indirect estimation introduces uncertainty, it represents a practical compromise given current technological constraints. Second, this study involved a limited sample of five healthy male participants. Although this enabled systematic validation under controlled conditions, the restricted sample size and gender homogeneity may limit generalizability. Future studies should include larger and more diverse populations to enhance the robustness of the framework. Third, the current analysis primarily focuses on ankle-related biomechanical adaptations, while potential compensatory effects on more proximal joints (e.g., knee and hip) and their associated muscles remain unexplored. Fourth, the simulation assumes an ideal rigid coupling between the exoskeleton and the ankle joint. In practice, the cuff, soft tissue, and Bowden cable introduce finite compliance that may affect load transfer, particularly during rapid torque transitions. This interface compliance is not explicitly modeled in the current framework. Future work may incorporate a simplified spring-damper representation to further improve simulation fidelity. Fifth, although the framework reasonably predicted the trends of biomechanical adaptations, the absolute precision of individual- and population-level predictions under both unassisted and assisted conditions requires further improvement. In particular, the simplification of the musculoskeletal model to the sagittal plane and the use of a single optimized control policy per condition contribute to the residual differences in waveform shape between simulation and experimental results. Future investigations may explore data-tracking trajectory optimization or EMG-informed control policy optimization to enhance predictive accuracy. Finally, the present study focused on a single assistance torque level determined by the experimental platform. Previous parameter-sweep studies suggest that plantar flexor effort generally decreases with higher assistance, possibly with diminishing returns [57, 58]. Since the proposed reflex-based controller adapts excitations through state-dependent feedback, similar trends can be expected. Future work will investigate a broader range of assistance magnitudes with paired experiments.

VI. CONCLUSION

This study proposes a predictive simulation framework for evaluating biomechanical adaptations during exoskeleton-assisted walking across different terrains, with particular emphasis on muscle effort (muscle activation and muscle force) responses. The effectiveness of the framework was systematically assessed through qualitative and quantitative analyses of the consistency between experimental and simulation results regarding kinematics, kinetics, muscle activation, and muscle force during level-ground, uphill, and downhill walking. The proposed predictive simulation framework demonstrated the

ability to reasonably predict the trends of changes in kinetics, plantar flexor muscle activation, and plantar flexor muscle force resulting from exoskeleton assistance across different terrains.

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